Ideas

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1 Architectures

1.1 RT-DETR

RT-DETR (Real-Time Detection Transformer) is a transformer-based object detector designed for real-time performance. Unlike traditional YOLO-style detectors, it removes the need for Non-Maximum Suppression (NMS) using set-based matching loss, similar to DETR [1]. RT-DETR variants incorporate efficient backbones and query mechanisms to balance speed and accuracy.

Key characteristics:

- End-to-end detection without NMS [1]
- Transformer-based object queries
- Designed for real-time GPU deployment
- Competitive accuracy–latency trade-off

papers:

- CARION et al., *End-to-End Object Detection with Transformers (DETR)*, ECCV 2020: `https://arxiv.org/abs/2005.12872`
- Liu et al., *RT-DETR: Real-Time Detection Transformer*, CVPR 2023: `https://arxiv.org/abs/2304.05049`

1.2 YOLO v26 (latest generation)

YOLO v26 represents the newest generation of the YOLO (You Only Look Once) family of one-stage object detectors. It follows the single-shot detection paradigm, predicting bounding boxes and class probabilities in a single forward pass [4]. Recent YOLO versions improve backbone efficiency and feature fusion for better real-time performance.

papers:

- Wong et al., *YOLO-NAS: Neural Architecture Search for YOLO*, 2023: `https://arxiv.org/abs/2302.00926`
- (YOLOv26 Ultralytics): `https://arxiv.org/html/2509.25164v1`

1.3 RF-DETR NAS

https://openreview.net/forum?id=qHm5GePxTh

1.4 Optimisations

To make architectures suitable for edge deployment, several optimisation techniques can be applied:

- **Quantisation:** Reducing numerical precision (e.g., FP32 to INT8) to decrease memory usage and improve inference speed
- **Structured pruning:** Removing entire channels or layers to reduce computational cost while preserving hardware efficiency [2].
- **Knowledge distillation:** Training a smaller student model to mimic a larger teacher model
- **Mixed precision inference:** Using different numerical precision levels across layers to balance speed and accuracy [3].
- **Hardware-aware optimisation:** Adapting model design to specific accelerators (e.g., NPUs, GPUs) and runtime compilers
- **Early exiting / dynamic inference:** Allowing models to stop computation early for simple inputs, reducing latency

papers:

- Jia et al., *Adaptive Distribution-aware Mixed-Precision Quantization*, 2025: `https://arxiv.org/abs/2510.19760`
- Zniber et al., *Hardware-aware NAS of Early Exiting Networks*, 2025: `https://arxiv.org/abs/2512.04705`
- Wu et al., *Energy-Efficient DNN Inference on Edge Devices*, MLSys 2023: `https://arxiv.org/abs/2303.12391`

References

[1] Nicolas Carion et al. End-to-end object detection with transformers. *ECCV*, 2020.

[2] Hao Li et al. Pruning filters for efficient convnets. *ICLR*, 2017.

[3] Paulius Micikevicius et al. Mixed precision training. *ICLR*, 2018.

[4] Joseph Redmon et al. You only look once: Unified, real-time object detection. *CVPR*, 2016.